PT MARKINAH

SPBG CNG untuk Angkutan Umum di Palembang

Basic Design and General Specification

B.6101. Instrument and Control Design Basis

0	11-10-2011	Issued for review			
Rev.	Date	Description	By	Checked	Appr.

1. OBJECTIVE

This document describes the Instrument and Control of SPBG (Stasiun Pengisian Bahan bakar Gas) Compressed Natural Gas (CNG) untuk Angkutan Umum di Palembang.

2. Electric transmission and Control

All transmitters and controlled final receivers be 4-20 mA and 24V.

All instrument will be installed with vent and drain for hazardous and/or polluting fluids the vent/drain or instrument shall be piped to the vent/ drain network.

Fire detection generally be based on analogue sensors, “smart” type, Hart Standard.

In buildings, use of addressable fire detectors, will be considered.

3. Station Monitoring System

Station monitoring system shall consist of interfacing module (PLC). Monitoring system shall be able to monitor the systems such as compressor system package.
Following requirement:

1. Configuration
   - Develop and modify application without process downtime.
2. Distributed logging
   - Logging and reporting of events
   - Maintain process history accurately
3. Distributed alarm and event
   - Alarming of critical event
   - Create an audit trail using event logging,
   - User friendly of visualization .

1. TUJUAN

Dokumen ini menjelaskan Instrumen dan Pengendalian SPBG (Stasiun Pengisian Bahan Bakar Gas) Compressed Natural Gas (CNG) untuk Angkutan Umum di Palembang.

2. Listrik transmisi dan Kontrol

Semua transmitters and controlled final receivers menggunakan 4-20 mA dan 24V.

Semua instrument akan diinstal ventilasi, selama ada polluting fluid berbahaya harus disalurkan ke jaringan ventilasi.

Deteksi kebakaran umumnya berdasarkan pada sensor analog, "Smart" Type, Hart Standar.

Dalam bangunan, akan dipertimbangkan penggunaan addressable untuk fire detector.

3. Sistem Stasion Monitoring/Pemantauan

Sistem stasiun pemantauan terdiri dari modul interfacing (PLC). Sistem Monitoring harus mampu memonitor sistem seperti memonitor system paket kompresor.
Persyaratan sebagai berikut:

1. Konfigurasi
   - Mengembangkan dan memodifikasi aplikasi tanpa harus proses downtime.
2. Distributed logging
   - Logging dan event report.
   - Maintain proses history secara akurat
3. Distributed alarm and event
   - Alarming of critical event
   - Membuat jejak audit menggunakan event log,
   - User friendly visualisasi.

4. Reporting
   For analyzing system data, station monitoring system shall provide a range of standard report, including:
   - Alarm/event report all alarm and event in a specified time period,
   - Alarm duration reports the time of occurrence and elapsed time before return to normal for specific alarm in specified time period.

4. Instrumentation for Package Unit

All instruments in packaged units shall be supplied fully piped and wired. All wiring or pneumatic tubing for remote transmission shall be terminated in junction boxes and/or the local panel at the package skid.

The relay logic system associated with the packaged unit may be installed locally close to the unit with the consideration of the proper protection for high radiation temperature and humidity act.

In principle, a common failure alarm shall be provided for a remote transmission. Alarm contacts shall be closed during plant normal operation.

4. Pelaporan
   Untuk menganalisis data system, stasiun monitoring system harus menyediakan berbagai laporan standar, termasuk:
   - Event alarm report semua alarm dan dalam periode waktu tertentu,
   - Durasi alarm report waktu kejadian dan waktu sampai kembali normal.

4. Instrumentasi untuk Package Unit

Semua instrumen dalam satu package sudah lengkap termasuk pemasangan pipa dan kabel. Semua kabel atau pneumatik untuk transmisi jarak jauh dipasangkan junction boxes atau local panel.

Logic relay yang berhubungan dengan Package unit dapat diinstal secara lokal dekat dengan unit, dengan pertimbangan perlindungan yang tepat terlindung dari suhu dan radiasi tinggi, kelembaban, hazardous dan lainnya.

Pada prinsipnya, a common failure alarm harus disediakan untuk transmisi jarak jauh. Alarm contact akan ditutup selama operasional normal.